WEEK 11 – PROJECT REPORT

AUTOMOBILE DATA CLEANING & TRANSFORMATION USING POWER BI

1. INTRODUCTION

Data cleaning and transformation are important steps in data analysis. Raw datasets may contain missing values, invalid data, and incorrect data types. In this project, the Automobile Dataset is cleaned and transformed using Microsoft Power BI. Power Query is used to prepare the data, and DAX is used to create measures for analysing automobile features such as price, horsepower, engine size, and fuel efficiency.

2. AIM

To clean, transform, and prepare the Automobile Dataset using Power BI for effective data analysis.

3. OBJECTIVES

- To identify and replace invalid values.
- To handle missing values.
- To change appropriate data types.
- To create calculated columns.
- To categorize vehicles based on their features.
- To create DAX measures for automobile analysis.

4. TOOLS USED

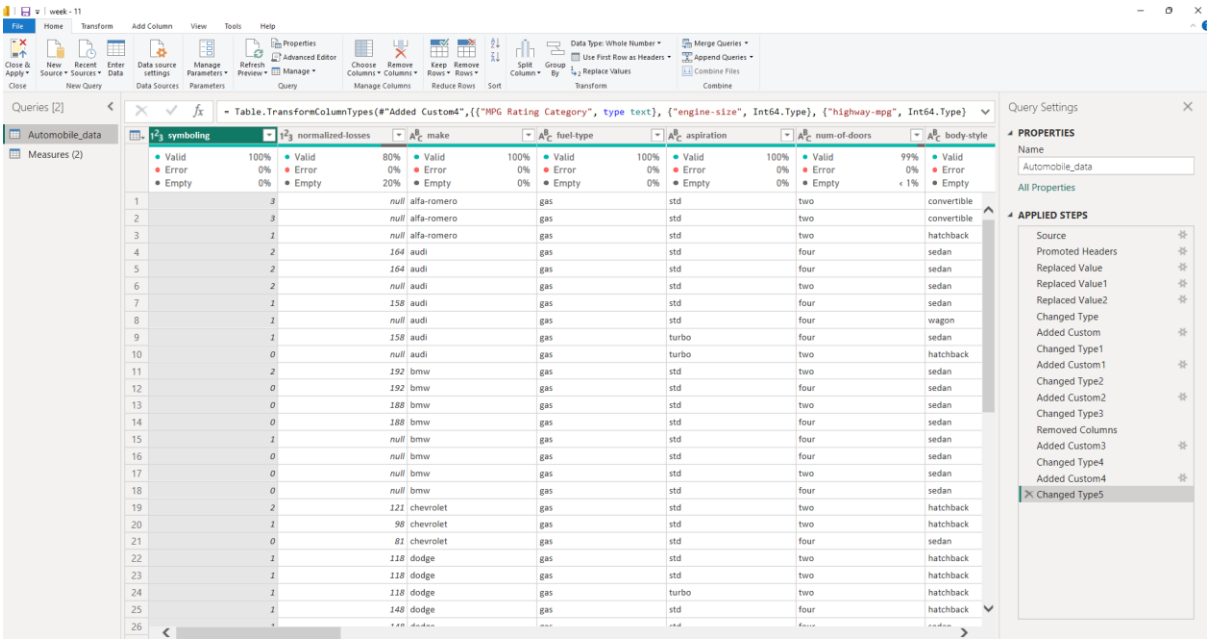
- Microsoft Power BI
- Power Query Editor
- DAX (Data Analysis Expressions)
- Automobile Dataset

5. DATA CLEANING

The following data cleaning operations were performed using Power Query:

1. Replaced invalid values (?) with null.
2. Identified and handled missing values.

3. Changed numerical columns into appropriate data types.
4. Cleaned and prepared the automobile data for analysis.



6. FEATURE ENGINEERING

Three new calculated columns were created.

1. Price per Horsepower

Formula: Price / Horsepower

Used to calculate the price per unit of engine horsepower.

2. Power-to-Weight Ratio

Formula: Horsepower / Curb-weight

Used to compare engine power with vehicle weight.

3. MPG Rating Category

Vehicles were classified based on highway mileage:

- Low – Below 20 MPG
- Medium – 20 to 29 MPG
- High – 30 MPG and above

1.2 Price per HorsePower	1.2 Power-to-Weight Ratio	MPG Rating Category
Valid 97%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%
Empty 3%	Empty 0%	Empty 0%
121.5765766	0.043563579	Medium
148.6486486	0.043563579	Medium
107.1428571	0.054551895	Medium
136.7647059	0.0436457	High
151.7391304	0.04072238	Medium
138.6363636	0.043877144	Medium
161	0.038677918	Medium
172	0.037237644	Medium
170.5357143	0.04536617	Medium
null	0.052407468	Medium
162.6732673	0.04217119	Medium
167.5742574	0.04217119	Medium
173.3057851	0.044649446	Medium
174.4214876	0.043761302	Medium
203.0165289	0.039607201	Medium
169.010989	0.056346749	Medium
227.0054945	0.053846154	Medium
202.6373626	0.05192582	Medium

7. VEHICLE CATEGORIZATION

Vehicles were categorized based on the following features:

- Body Style
- Drive Wheels
- Fuel System

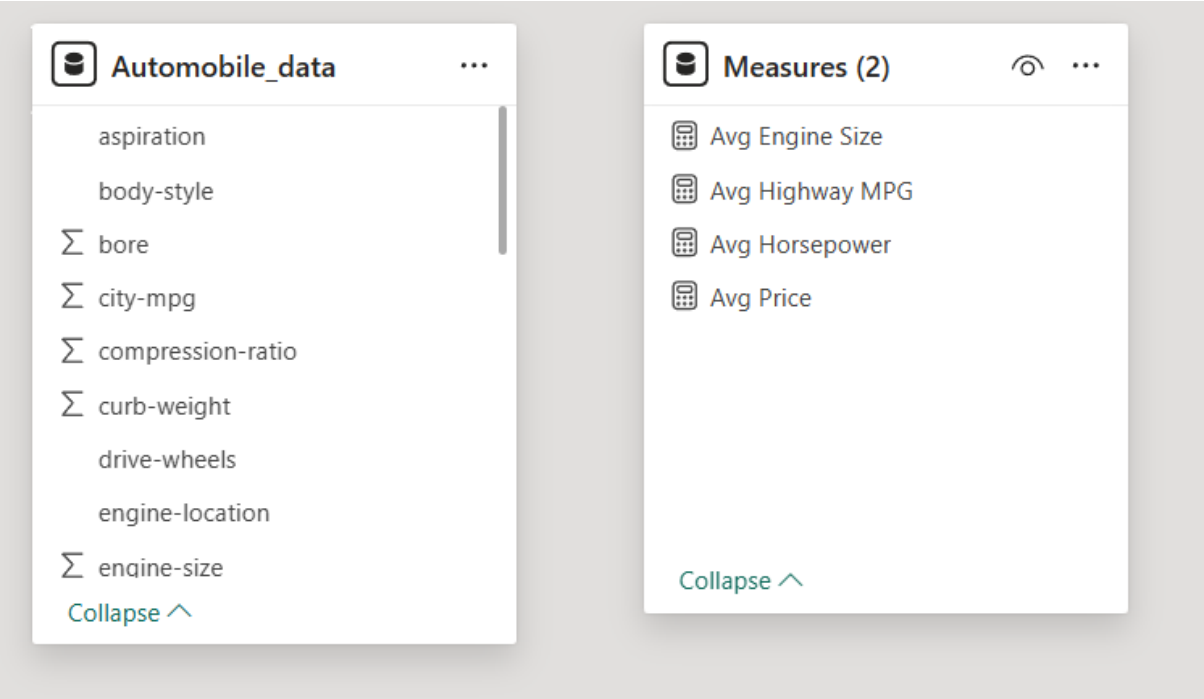
This categorization helps compare different types of automobiles.

8. DAX MEASURES

Four DAX measures were created for automobile analysis:

1. Average Price
2. Average Horsepower
3. Average Engine Size
4. Average Highway MPG

These measures help analyse vehicle pricing, engine performance, and fuel efficiency.



9. RESULT

The Automobile Dataset was cleaned and transformed using Power BI. New calculated columns, vehicle categories, and DAX measures were prepared for further analysis.

10. CONCLUSION

The Automobile Data Cleaning and Transformation project was carried out using Power Query and DAX in Power BI. The prepared dataset provides useful information about automobile pricing, engine performance, and fuel efficiency. This dataset can be used to develop an interactive Automobile Analytics Dashboard in the next stage.